

Exploration in RL: Multi-armed Bandit (Continue)

Kianté Brantley
CS 1840 Fall 2026

Announcements

HW1 is out, and it's due today (**09/28 @ 12 midnight**)!

HW2 released last Friday!

Slight relaxation of HW late hour policy (**please check course website**)

My office hours are **canceled** today.

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Recap: MAB

Interactive learning process:

For $t = 0 \rightarrow T - 1$

(# based on historical information)

1. Learner pulls arm $a_t \in \{1, \dots, K\}$
2. Learner observes an i.i.d reward $r_t \sim \nu_{a_t}$ of arm a_t

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Learning metric:

$$\text{Regret}_T = T\mu^\star - \sum_{t=0}^{T-1} \mu_{a_t}$$

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For $k = 1 \rightarrow K$: (# Exploration phase)

Pull arm- k N times, observe $\{r_i\}_{i=1}^N \sim \nu_k$

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For $t = NK \rightarrow T - 1$: (# Exploitation phase)

Pull the best empirical arm, i.e., $a_t = \arg \max_{i \in [K]} \hat{\mu}_i$

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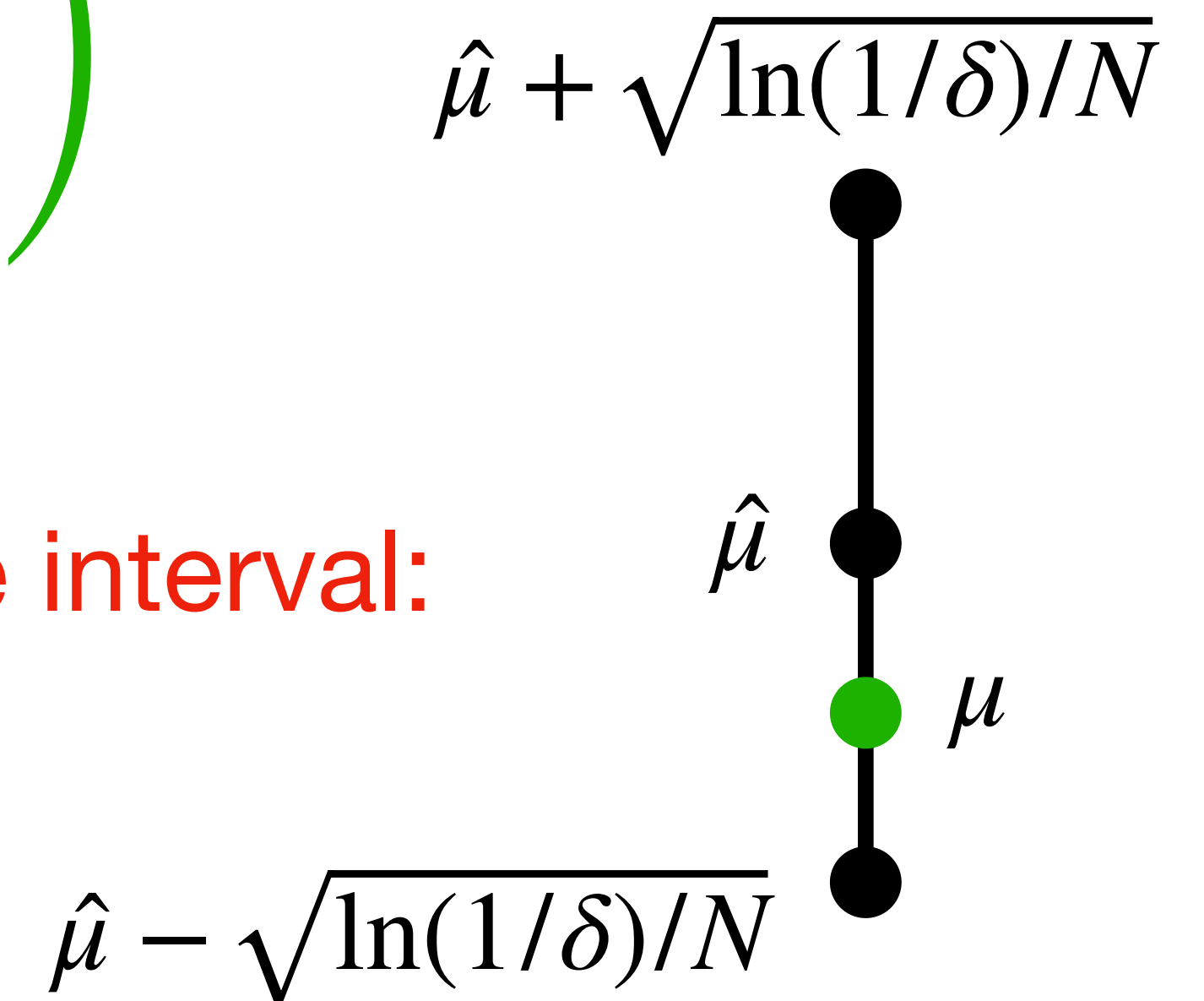
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After the Exploration phase, with probability at least $1-\delta$, **for all**
arm $i \in [K]$, we have:

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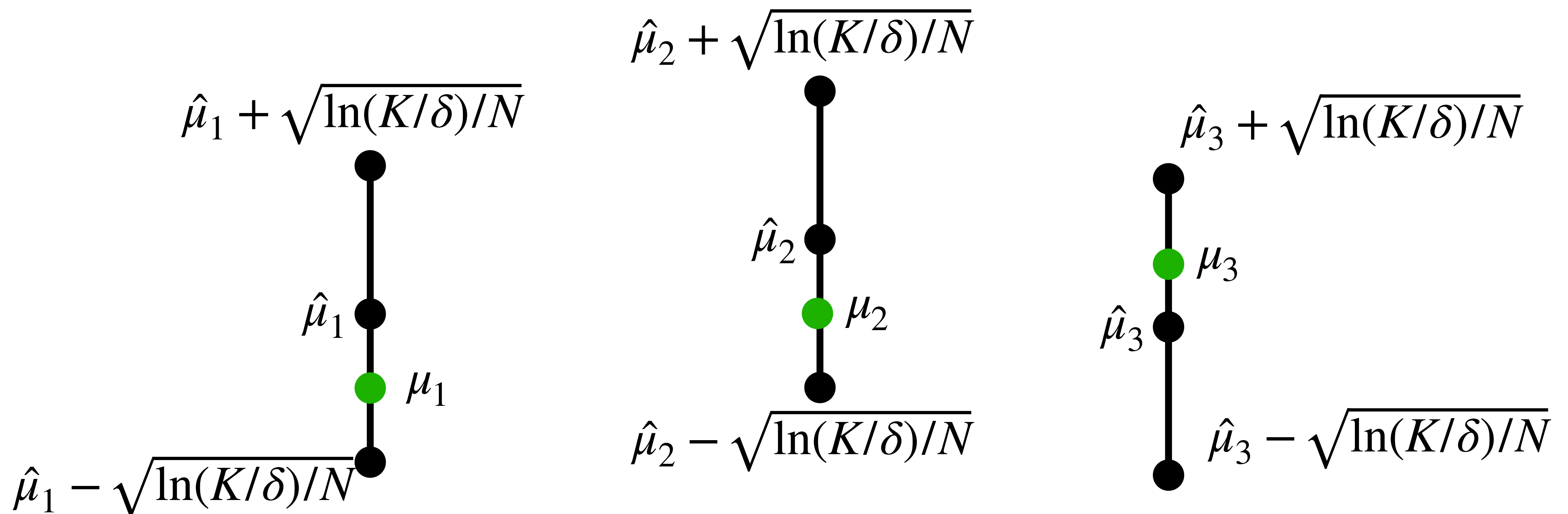
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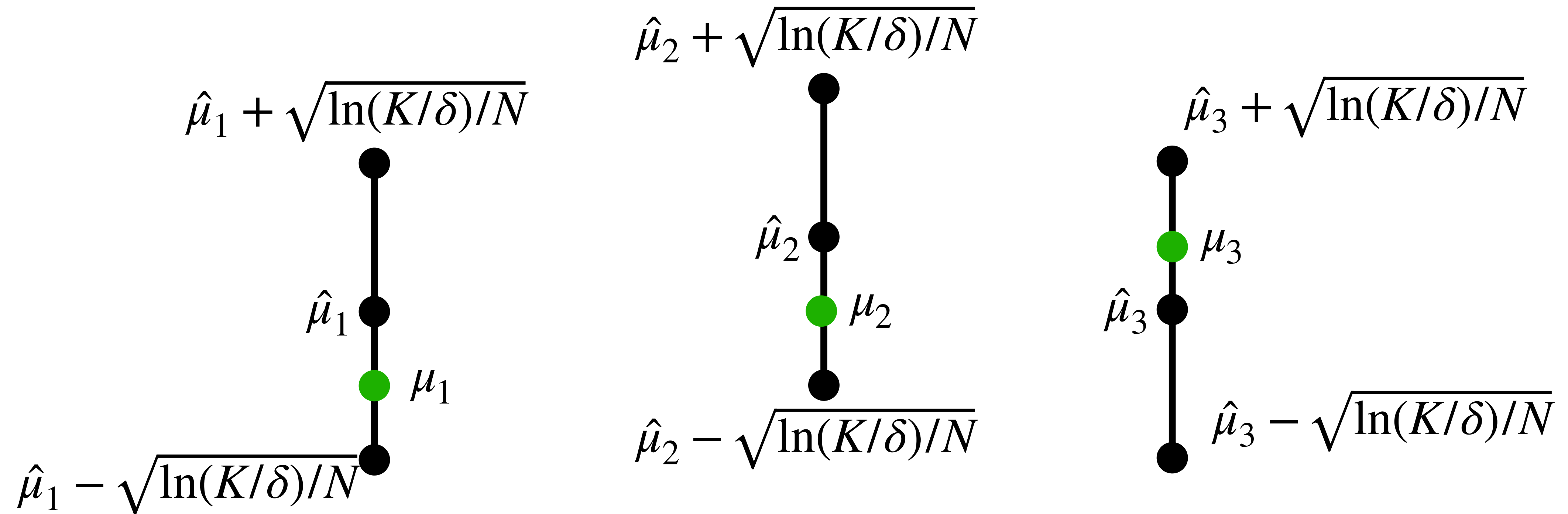
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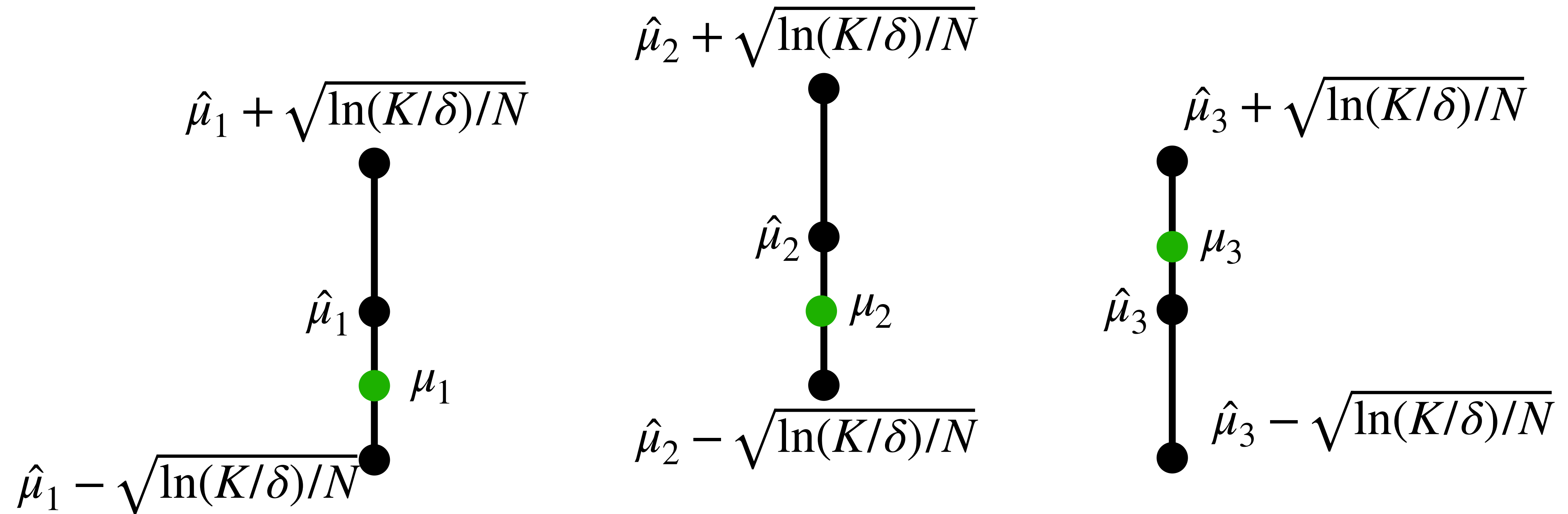
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In the rest, we will condition on the event that the confidence intervals are valid...



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Recall the Alg in this case will pick $a_t = 2$, for all $t \geq NK$,
 (but it will suffer regret $(T - NK)(\mu_3 - \mu_2)$)

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Denote empirical best arm $\hat{k} = \arg \max_{i \in [K]} \hat{\mu}_i$, and THE best arm $k^\star = \arg \max_{i \in [K]} \mu_i$

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Let's now bound $\text{Regret}_{\text{exploit}}$

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$$\leq 2\sqrt{\ln(K/\delta)/N}$$

$$\text{Regret}_{\text{exploit}} \leq (T - NK)(\mu_{k^\star} - \mu_{\hat{k}}) \leq 2T\sqrt{\frac{\ln(K/\delta)}{N}}$$

Finally, combine two regret together:

$$\text{Regret}_{\text{explore}} \leq N(K - 1) \leq NK$$

$$\text{Regret}_{\text{exploit}} \leq (T - NK)(\mu_{k^*} - \mu_{\hat{k}}) \leq 2T\sqrt{\frac{\ln(K/\delta)}{N}}$$

$$\text{Regret}_T = \text{Regret}_{\text{explore}} + \text{Regret}_{\text{exploit}} \leq NK + 2T\sqrt{\frac{\ln(K/\delta)}{N}}$$

Minimize the upper bound via optimizing N:

$$\text{Set } N = \left(\frac{T\sqrt{\ln(K/\delta)}}{K} \right)^{2/3}, \text{ we have:}$$

$$\text{Regret}_T \leq O\left(T^{2/3}K^{1/3} \cdot \ln^{1/3}(K/\delta)\right)$$

To conclude:

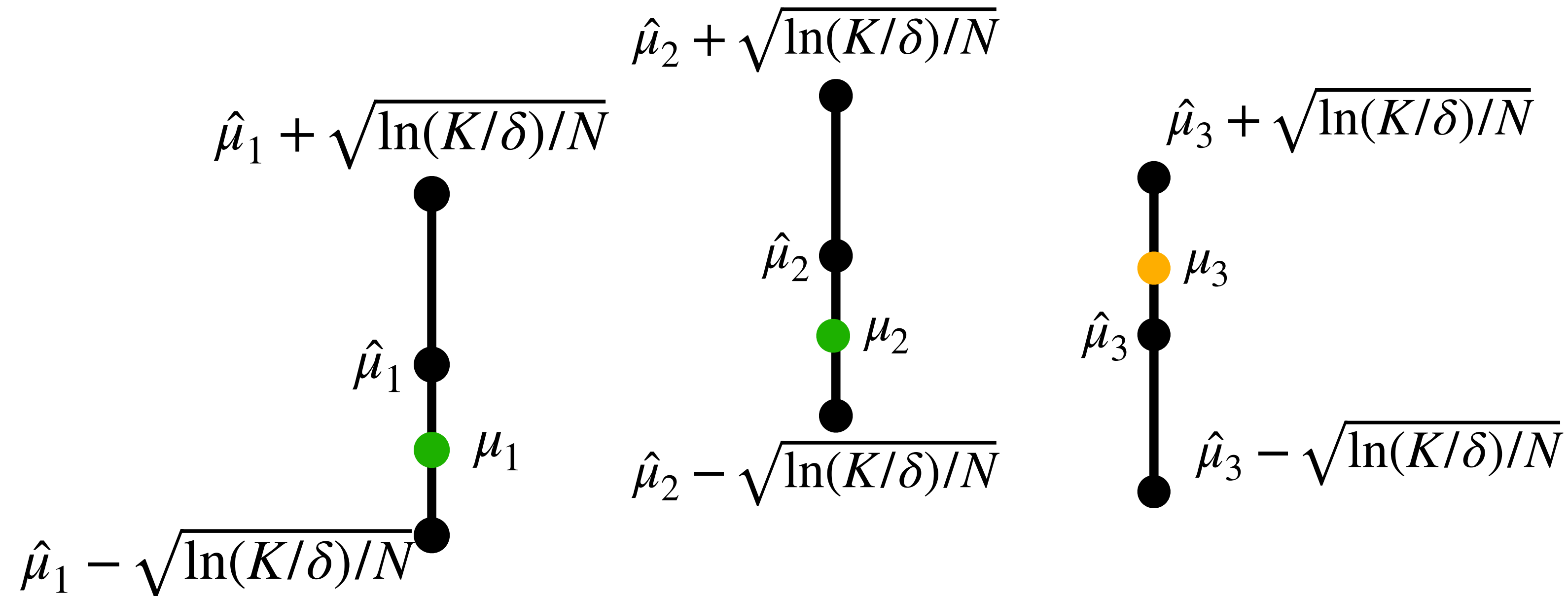
[Theorem] Fix $\delta \in (0,1)$, set $N = \left(\frac{T\sqrt{\ln(K/\delta)}}{2K} \right)^{2/3}$, with probability at least $1 - \delta$, **Explore-Then-Commit** has the following regret:

$$\text{Regret}_T \leq O \left(T^{2/3} K^{1/3} \cdot \ln^{1/3}(K/\delta) \right)$$

(See the MAB reading material for more details)

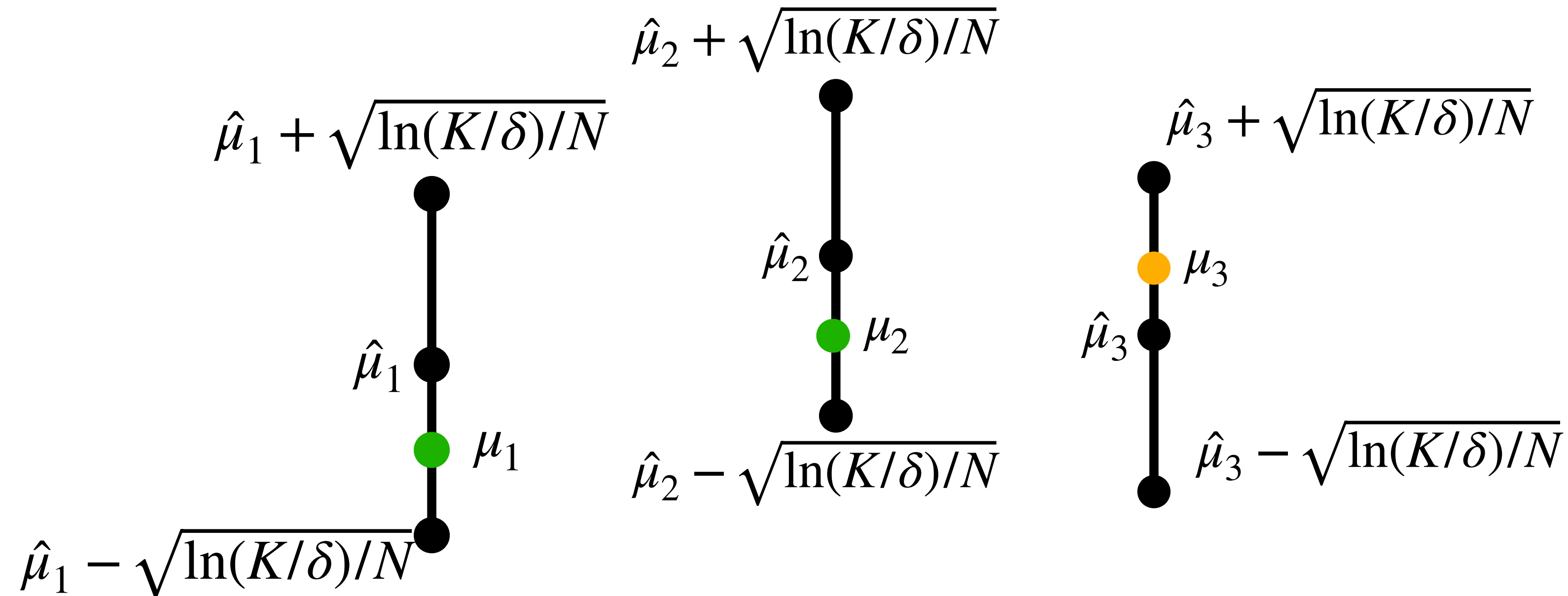
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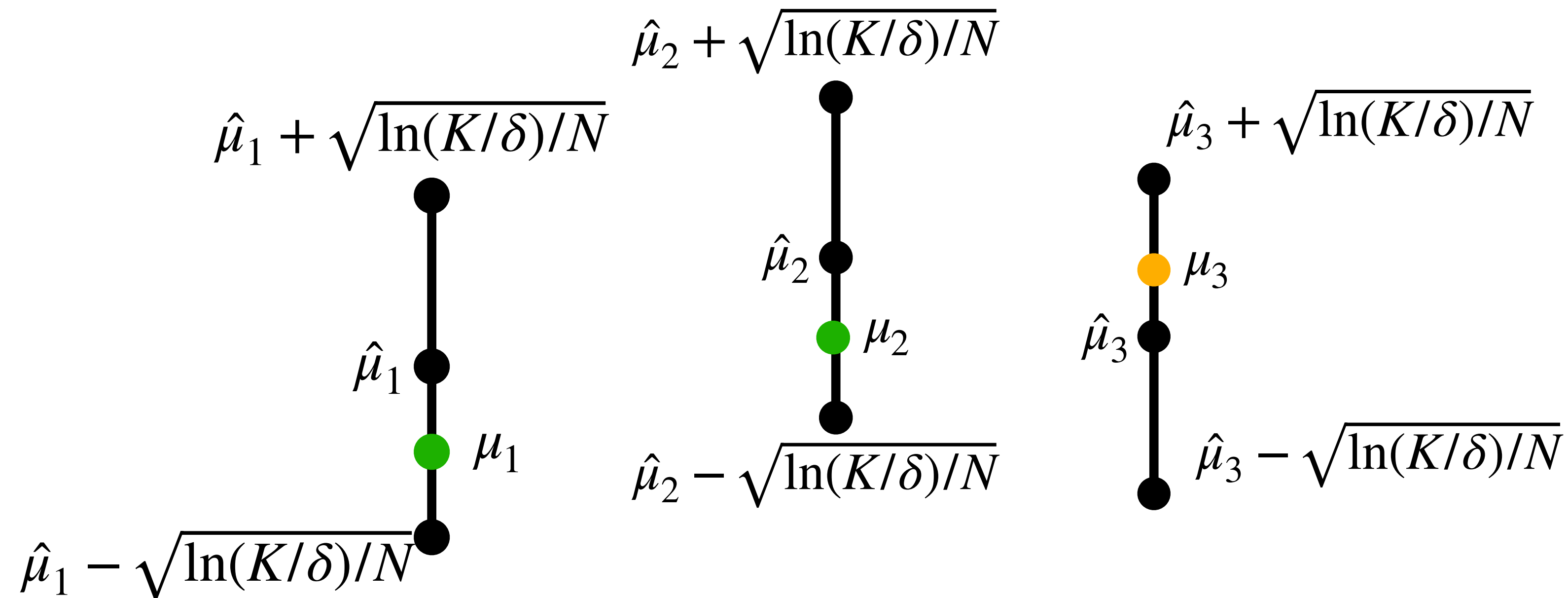
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But from the picture, we see that $(\mu_3 - \mu_2) \leq \text{length-of-Confidence-Interval}$

Question for Today:

Can we design an algorithm that achieves $\widetilde{O}(\sqrt{T})$ regret?

Outline:

1. The upper Confidence Bound Algorithm
2. Analysis of UCB algorithm

Statistics that we maintain during learning:

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$$\text{i.e., } \hat{\mu}_t(i) = \sum_{\tau=0}^{t-1} \mathbf{1}\{a_\tau = i\} r_\tau / N_t(i)$$

Recall the Tool for Building Confidence Interval:

[Hoeffding] Given a distribution $\nu \in \Delta([0,1])$, and N i.i.d samples $\{r_i\}_{i=1}^N \sim \nu$, w/ probability at least $1 - \delta$, we have:

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Thus, we know that for all iteration t , we have the for all $i \in [K]$, w/ prob $1 - \delta$,

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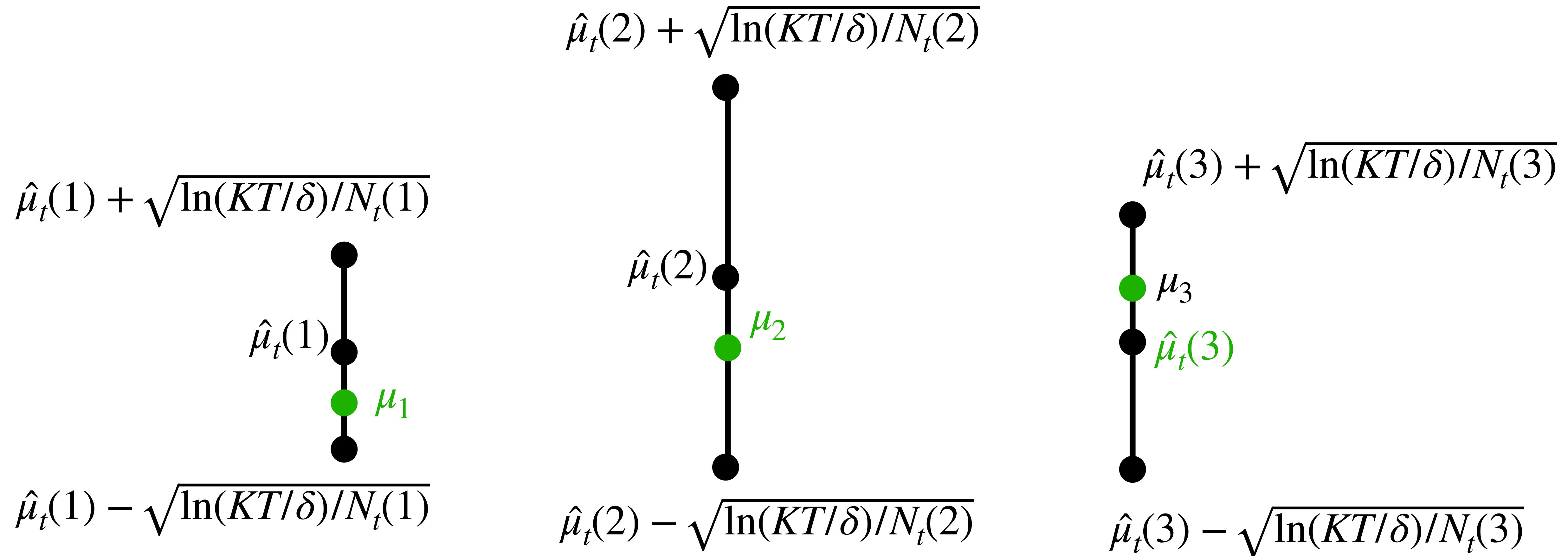
(Note that I applied union bound over all $t \in [T]$ and all $i \in [K]$, but let's not worry too much about log terms—*details are in reading material in case you are interested*)

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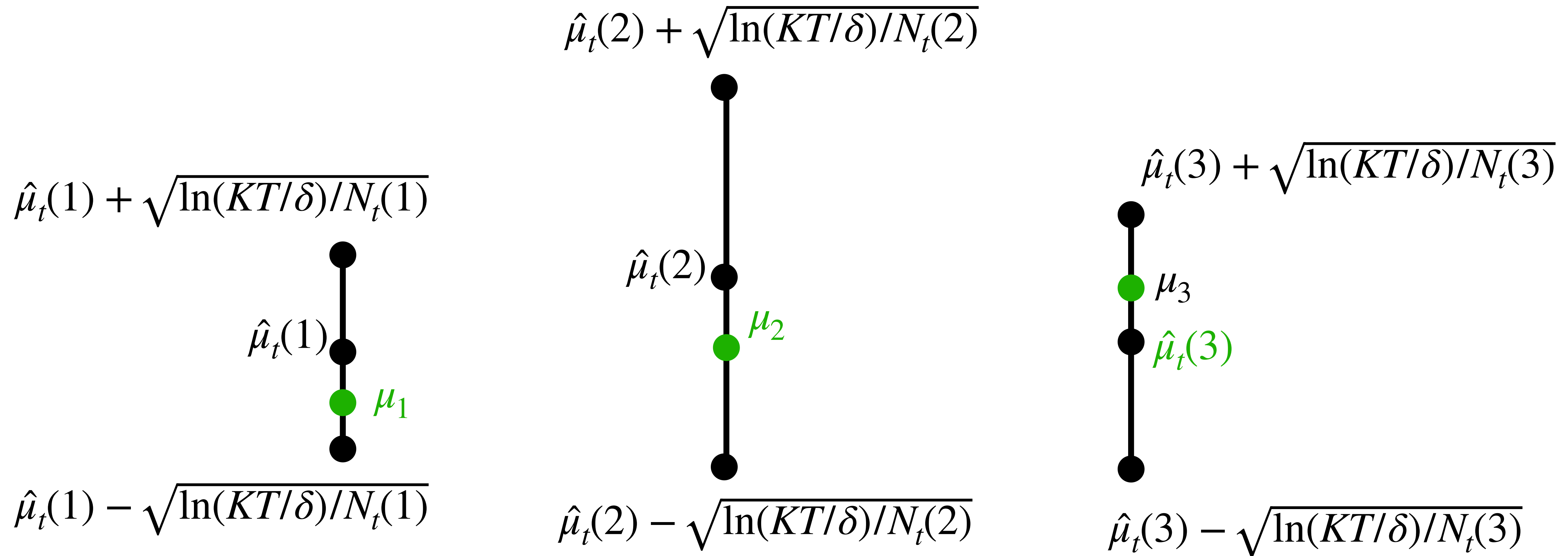


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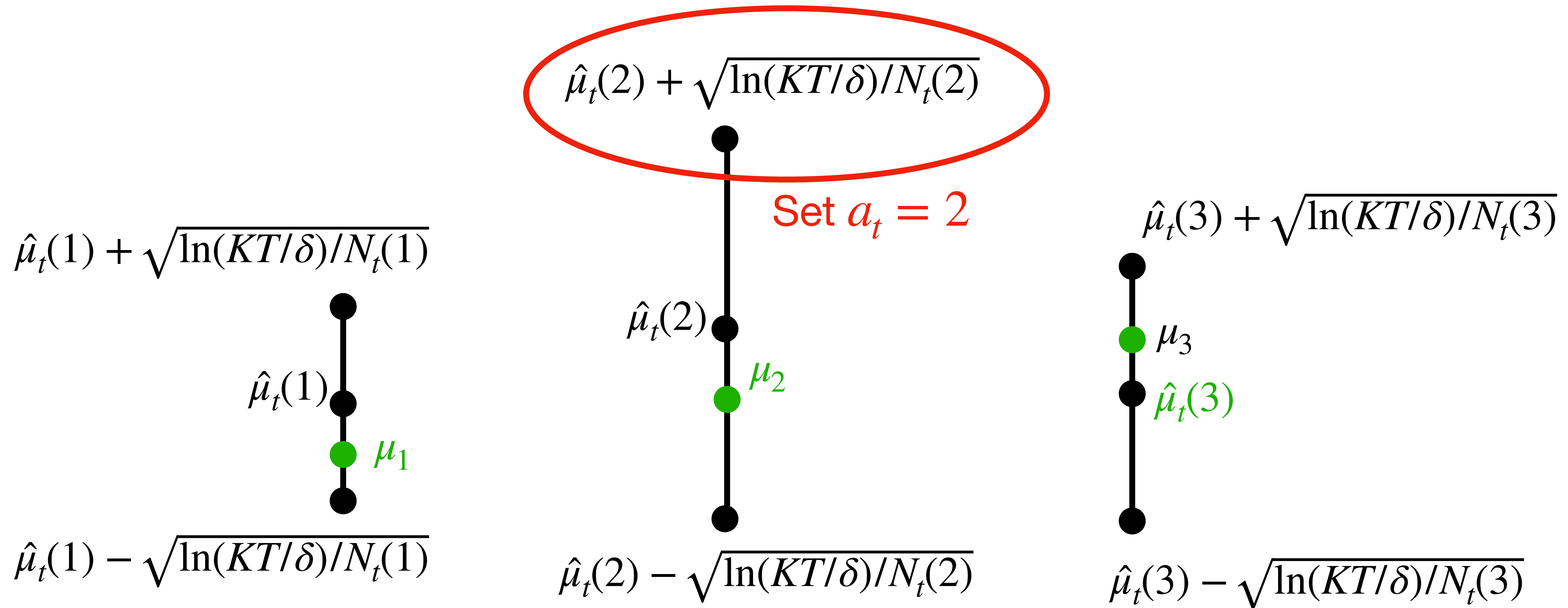
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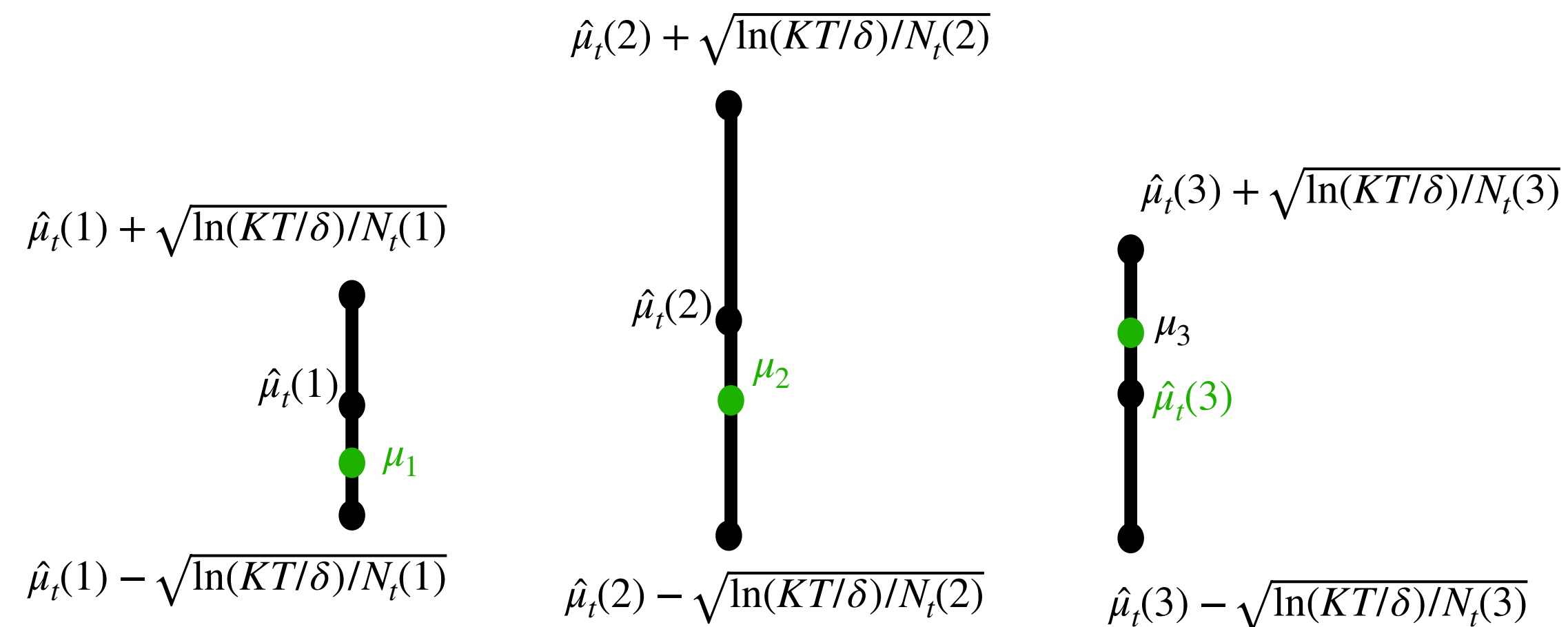
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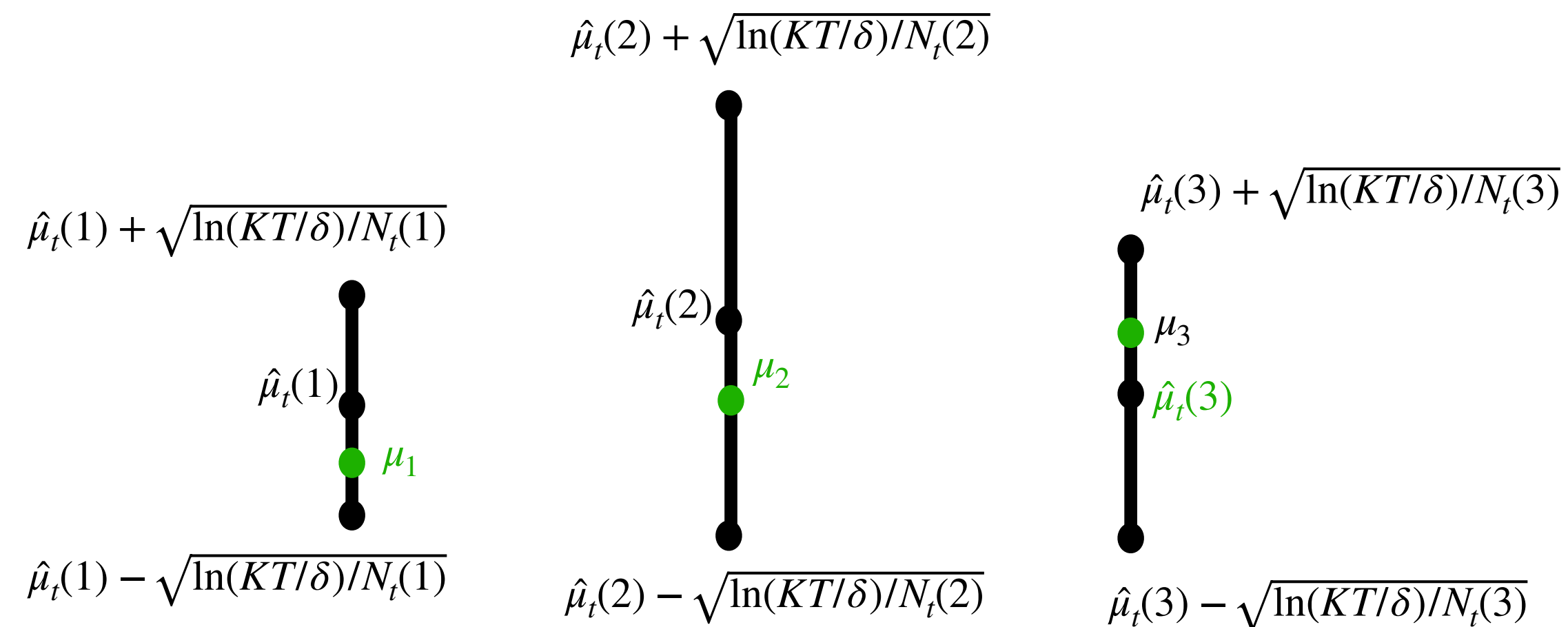


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“Reward Bonus”: $\sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

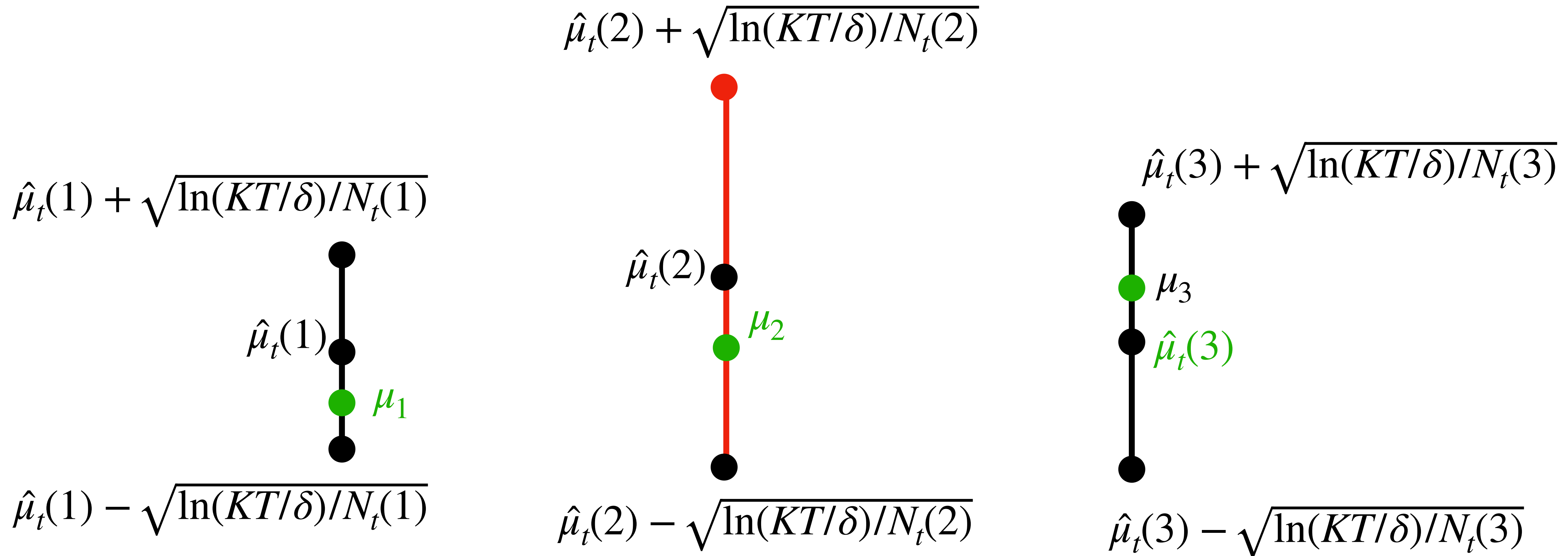
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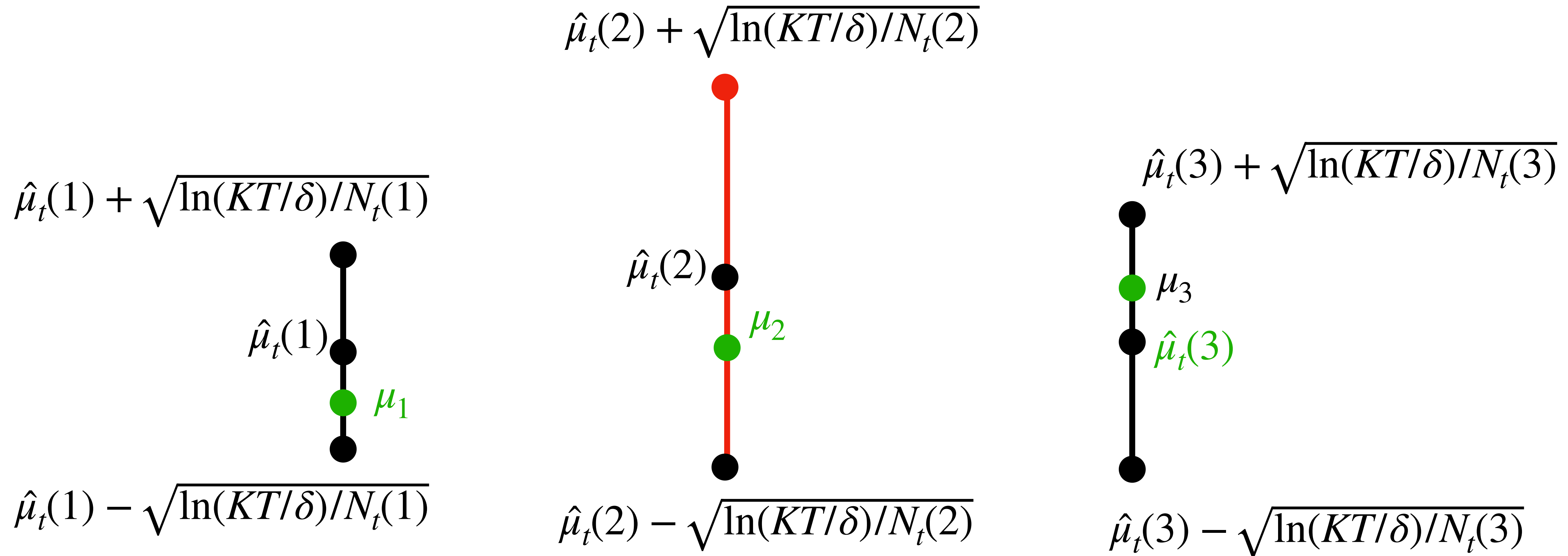
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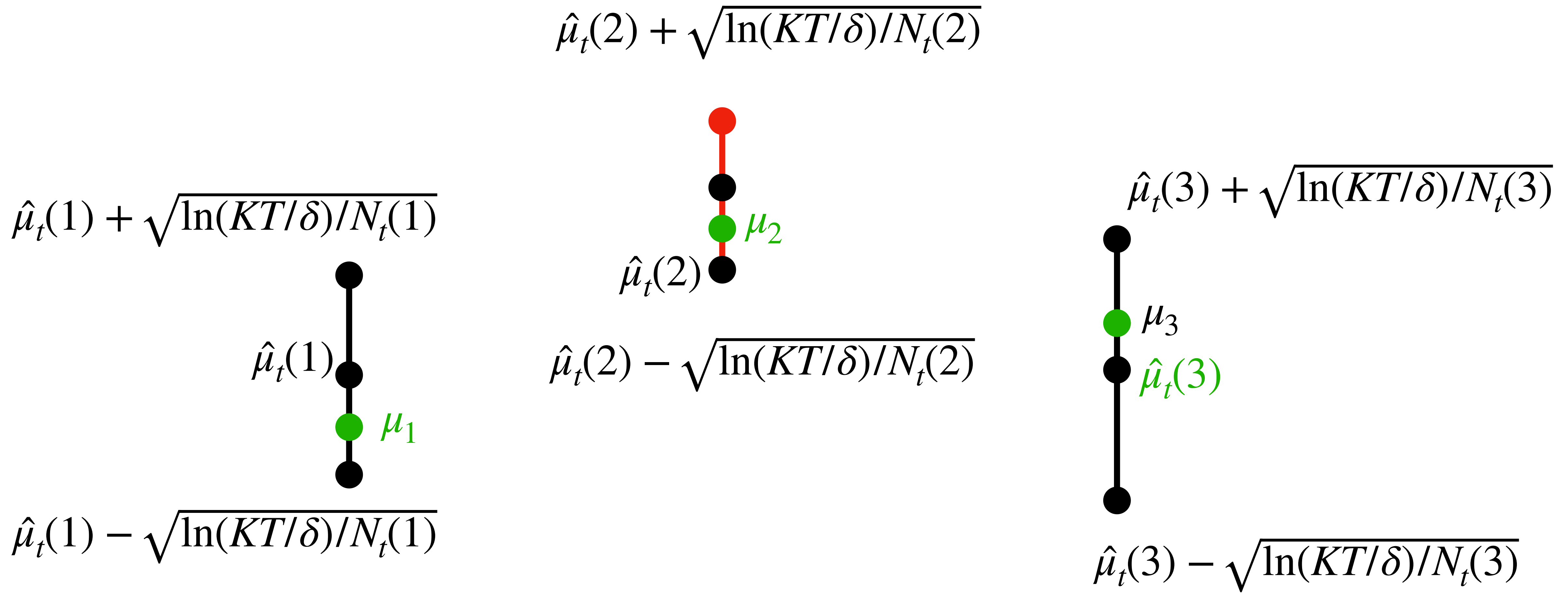
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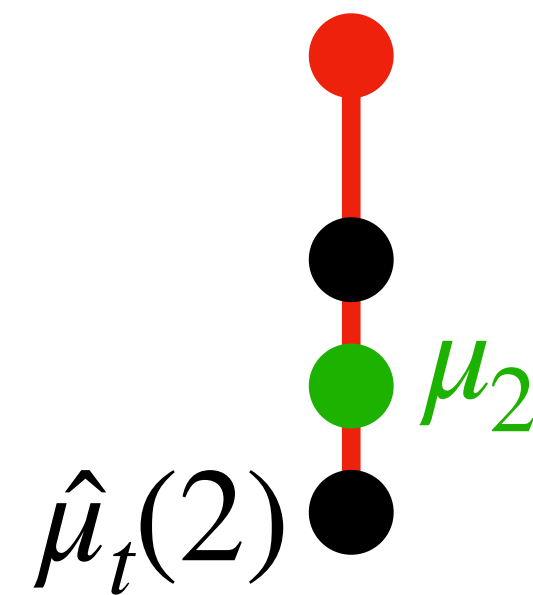
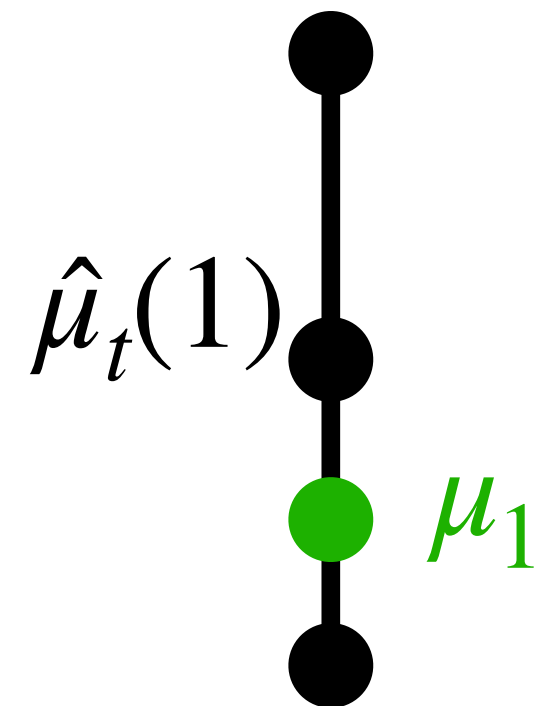


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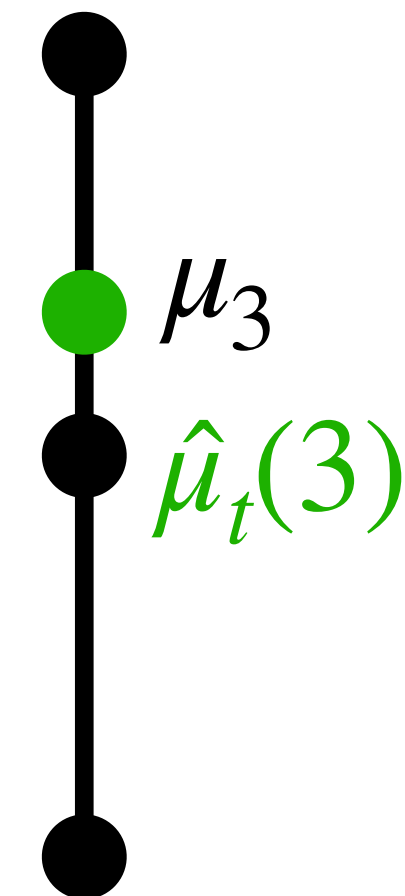
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Case 2: a_t has small conf-interval, then it is simply a good arm, i.e., its true mean is pretty high!

Thus, we do exploitation in this case!

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

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$$\begin{aligned} \text{Regret-at-}t &= \mu^\star - \mu_{a_t} \\ &\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t} \end{aligned}$$

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$$\leq 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}}$$

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

$$\text{Regret-at-}t = \mu^\star - \mu_{a_t}$$

Q: why?

$$\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t}$$

$$\leq 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}}$$

Case 1: $N_t(a_t)$ is small
(i.e., uncertainty about a_t is large);

We pay regret, BUT we **explore** here,
as we just tried a_t at iter t !

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

$$\begin{aligned} \text{Regret-at-}t &= \mu^\star - \mu_{a_t} \\ &\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t} \\ &\leq 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} \end{aligned}$$

Case 2: $N_t(a_t)$ is large, i.e., conf-interval of a_t is small,

Then we **exploit** here, as a_t is pretty good (the gap between $\mu^\star$ & μ_{a_t} is small)!

Let's formalize the intuition

Finally, let's add all per-iter regret together:

$$\begin{aligned}\text{Regret}_T &= \sum_{t=0}^{T-1} \left(\mu^\star - \mu_{a_t} \right) \\ &\leq \sum_{t=0}^{T-1} 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} \\ &\leq 2\sqrt{\ln(TK/\delta)} \cdot \sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(a_t)}}\end{aligned}$$

Let's formalize the intuition

Finally, let's add all per-iter regret together:

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Lemma (optional):

$$\sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(a_t)}} \leq O\left(\sqrt{KT}\right)$$

UCB Regret:

[Theorem (informal)] With high probability, UCB has the following regret:

$$\text{Regret}_T = \widetilde{O} \left(\sqrt{KT} \right)$$

(See reading material for more details)

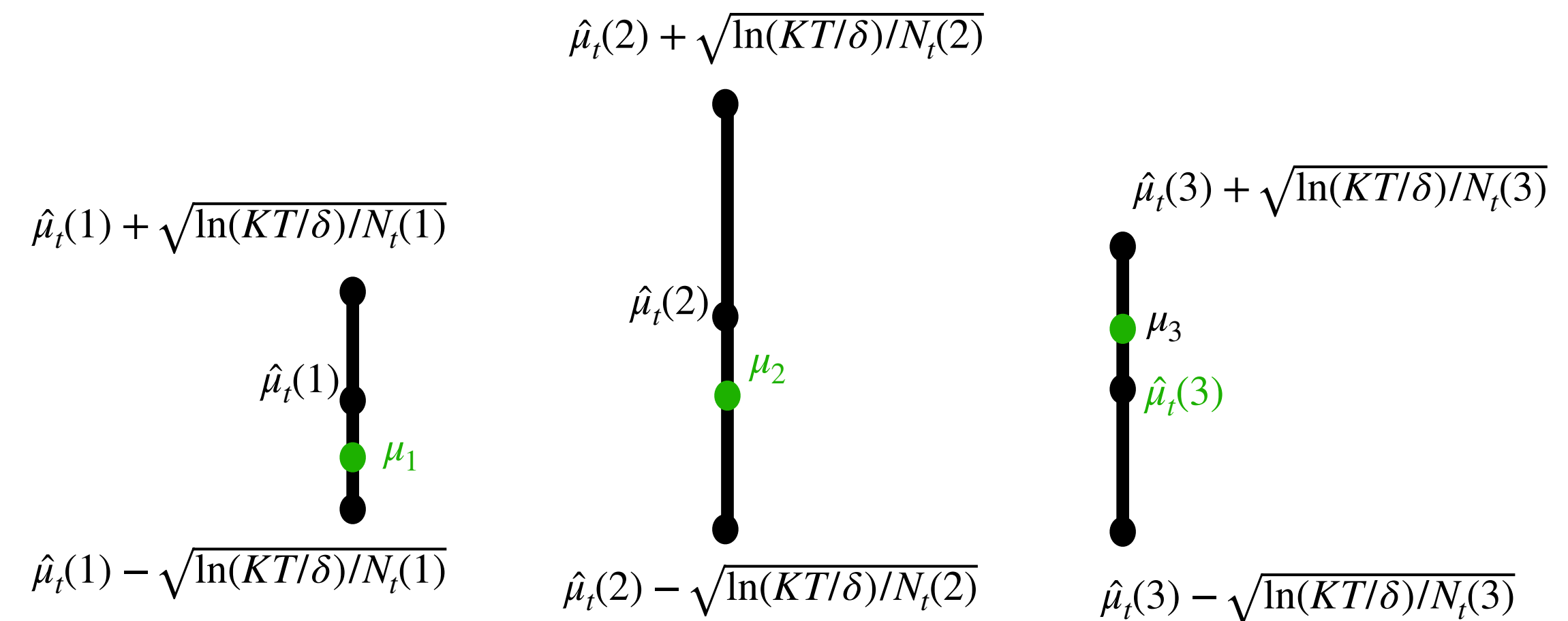
Summary for Today:

UCB algorithm: *Principle of Optimism in the face of Uncertainty*

For $t = 0 \rightarrow T - 1$:

$$a_t = \arg \max_{i \in [K]} \left(\hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}} \right)$$

(# Upper-conf-bound of arm i)



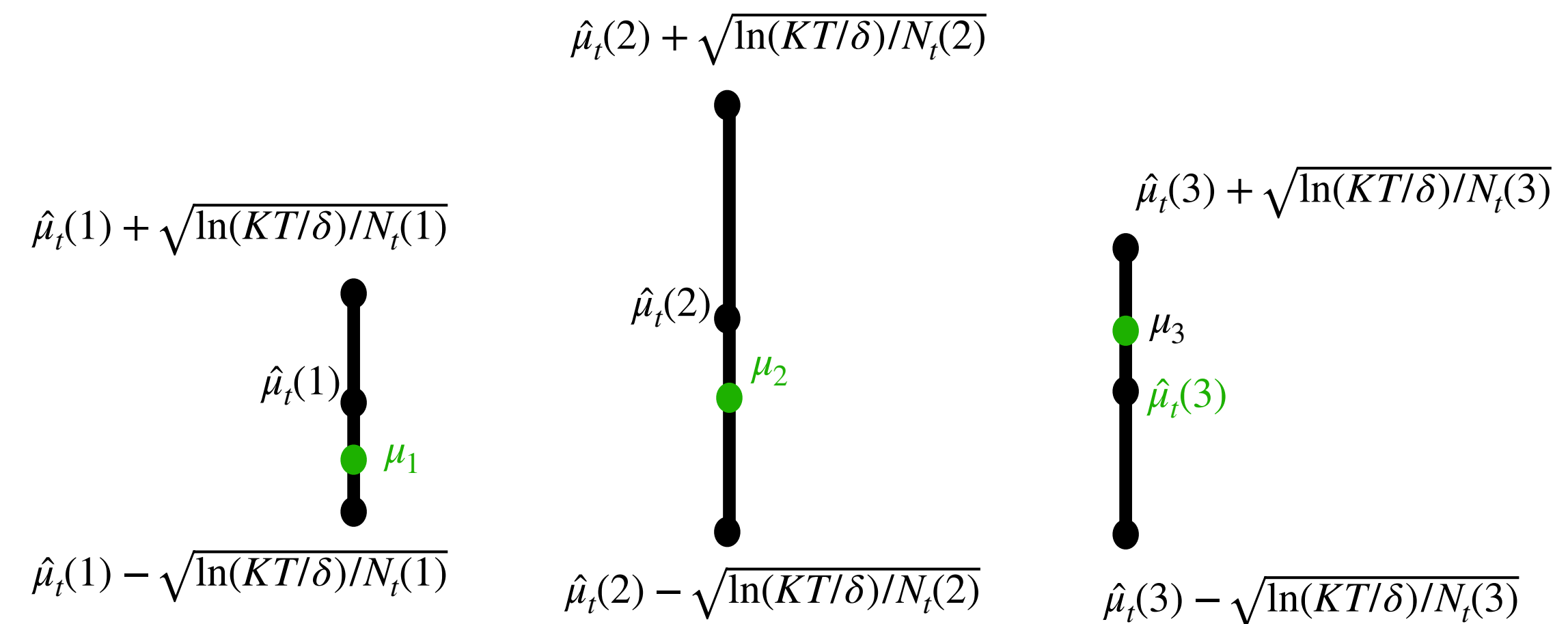
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Analysis Intuition:

Case 1: the arm a_t has high uncertainty (we explore)

Case 2: the arm a_t has low uncertainty, then it must be a near-optimal arm (i.e., exploit)

Feedback and Attendance

<https://bit.ly/4gYn2nC>

